

Arpit Mathur

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RESEARCH INTERESTS

Human-AI interaction, data visualization, and clinical decision support. I design and evaluate interpretable, AI-driven visual analytics systems that help clinicians, researchers, and patients make sense of complex health and scientific data.

EDUCATION

Carnegie Mellon University

Ph.D. - Human-Computer Interaction

Advisor: Adam Perer

Pittsburgh, PA

Aug 2024 - Present

Georgia Institute of Technology

Master of Science - Human-Computer Interaction

Atlanta, GA

Aug 2019 - May 2021

Indian Institute of Technology, Guwahati

Bachelor of Design - Design

Minor in Mathematics

Guwahati, India

Aug 2015 - May 2019

PUBLICATIONS

Under Review

A. Mathur*, C. Klinhom*, A. Patel, K. Morrison, J. Lai, P. Correa-Jaque, S. W. Lee, S. Lin, M. Griffiths, M. Kanwar, R. Benza, Y. Matusov†, and A. Perer†. 2026. ourPHORA: The Risks and Rewards of Patient-Facing AI Survival Assessment for Pulmonary Arterial Hypertension. *Under review*.

S. Bali*, **A. Mathur***, M. Goel†, and A. Perer†. 2026. AXES: Co-Constructing Semantic Dimensions for Literature Exploration. *Under review*.

E. Mason, **A. Mathur**, T. Chen, E. Wu, V. Sivaraman, K. Morrison, J. Kahn, C. McQuade, D. DiNardo, and A. Perer. 2026. Framing Substitution and Proxy Confidence: A Qualitative Study of Medical Students' LLM Use During Clerkships. *Under review*.

K. Kim, **A. Mathur**, A. Perer, and J. Seo. 2026. Vital Scape: Visual Analytics for Interpreting Physiological Foundation Model. *Under Review*.

W. Epperson, **A. Mathur**, D. Moritz, and A. Perer. Texture: Structured Exploration of Text Datasets. *Under review*.

Conference Papers

Morrison, K., **Mathur, A.**, Bradshaw, A., Wartmann, T., Lundi, S., Zandifar, A., Dai, W., Batmanghelich, K., Eslami, M., & Perer, A. (2025). [A Human-Centered Approach to Identifying Promises, Risks, & Challenges of Text-to-Image Generative AI in Radiology](#). Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, 8(2), 1758-1770.

Ben Rydal Shapiro, Rogers Hall, **Arpit Mathur**, and Edwin Zhao. 2025. [Exploratory Visual Analysis of Transcripts for Interaction Analysis in Human-Computer Interaction](#). In CHI Conference on Human Factors in Computing Systems (CHI '25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 17 pages.

Bradshaw, A., Morrison, K., **Mathur, A.**, Dai, W., Eslami, M., Batmanghelich, K., & Perer, A. (2025). [Toward Interpretable 3D Diffusion in Radiology: Token-Wise Attribution for Text-to-CT Synthesis](#). In Medical Imaging with Deep Learning-Short Papers.

Shapiro, B. R., Hall, R., **Mathur, A.**, & Zhao, E. (2024). [Turn Charts for Interaction Analysis: Visually Mapping the Conversation Floor](#). In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024 (pp. 43-50). International Society of the Learning Sciences.

Mathur, A. & Shapiro, B.R. (2022). [Interactive Transcription Techniques for Interaction Analysis](#). In Proceedings of the 16th International Conference of the Learning Sciences (ICLS). International Society of the Learning Sciences, pg. 19-26. Hiroshima, Japan: International Society of the Learning Sciences.

Fulfagar, L., Gupta, A., **Mathur, A.**, Shrivastava, A. (2021). [Development and Evaluation of Usability Heuristics for Voice User Interfaces](#). In: Chakrabarti, A., Poovaiah, R., Bokil, P., Kant, V. (eds) Design for Tomorrow—Volume 1. ICoRD 2021. Smart Innovation, Systems and Technologies, vol 221. Springer, Singapore.

Mathur, A., Bhowmick, S., Sorathia, K. (2019). [A Study of Outbound Automated Call Preferences for DOTS Adherence in Rural India](#). In: Lamas, D., Loizides, F., Nacke, L., Petrie, H., Winckler, M., Zaphiris, P. (eds) Human-Computer Interaction – INTERACT 2019. INTERACT 2019. Lecture Notes in Computer Science(), vol 11748. Springer, Cham.

Workshop Papers

Eric Mason, Truanne Chen, **Arpit Mathur**, Katelyn Morrison, Ethan Wu, and Adam Perer. 2026. [AI in Clinical Training: Examining Medical Student LLM Use Through a Clinical Sensemaking Lens](#). In Proceedings of CHI '26 Workshop on Sensemaking.

Mathur, A., Perer, A. (2025). [Clini-Compare: An Interactive Patient-Similarity Visualization Tool for Clinical Decision Support](#). In: Avellino, I., Kuo, P.-Y., Foong, P. S., Wiese, J., Mentis, H., Munson, S., Wallace, J., Singh, A., Miller, A., Epstein, D., & Nunes, F. (2025, April 12). CHI '25 Workshop on Envisioning the Future of Interactive Health.

Poster Presentations

Mathur, A., Bruce, C. (2021). [A Study of Motivation, Preferences, and Pain Points Regarding Participation in Career Related Mentorship](#). In: Stephanidis, C., Antona, M., Ntoa, S. (eds) HCI International 2021 - Posters. HCII 2021. Communications in Computer and Information Science, vol 1419. Springer, Cham.

RESEARCH EXPERIENCE

Carnegie Mellon University, HCI Institute

Pittsburgh, PA

Doctoral Research Assistant

Aug 2024 - Present

- ourPHORA (patient-facing AI survival assessment for pulmonary arterial hypertension). Led formative interviews with 18 patients, caregivers, and support group leaders. Designed an interactive patient-facing dashboard and evaluated it in a mixed-methods study with 44 patients and caregivers. Now running an in-clinic study with patients and clinicians at Cedars-Sinai Medical Center.
- AXES (mixed-initiative literature analysis). Designed and built a visualization system (SvelteKit, D3, LLM/RAG pipeline) in which researchers and AI co-construct the semantic dimensions used to compare papers. Ran a within-subjects mixed-methods study with 15 researchers.
- cBioPortal (LLM augmented visual data analysis). Designing LLM integrations for the cBioPortal cancer genomics dashboard in collaboration with Memorial Sloan Kettering Cancer Center. Designing a 2×2 mixed factorial study crossing input and output modality, and developing interactive prototypes for each cell to facilitate the study.
- Clini-Compare (patient similarity for decision support). Built a patient-similarity visualization tool that retrieves semantically similar patients from MIMIC-IV radiology notes using sentence embeddings, with a UMAP overview, similarity histogram, and side-by-side note comparison.

Georgia Institute of Technology

Atlanta, GA

Graduate Research Assistant

Jan 2020 - May 2021

- Designed and Developed Interactive web based visualizations as course material for EC 3040 Semiconductor Devices.
- 4 interactive chapters were used by 100 students enrolled in EC 3040 over the fall semester 2020, to facilitate remote learning.

Indian Institute of Technology, Guwahati

Guwahati, India

Undergraduate Research Assistant.

Aug 2018 - May 2019

- Worked with the Govt. of Assam to Design and Test an IVR based Information Dissemination solution in 3 Rural districts of Assam, with 2000+ Patients.
- The IVR system was used to measure and improve adherence to the Tuberculosis (TB) Directly Observed Therapy (DOTS).

INDUSTRY EXPERIENCE

The MathWorks

Natick, MA

User Experience Designer

Jun 2021 - Jun 2024

- Supported multiple MATLAB and Simulink products, including MATLAB fixed-point workflows, Simulink Studio, and System Composer.
- Led 10+ collaborative design events, developed wireframes, and iteratively prototyped high-fidelity designs using Figma.

SAS Research and Development

Pune, India

User Experience Design Intern

May 2018 - Jul 2018

- Redesigned the SAS Analytics Credit Scoring Tool. Conducted Contextual Interviews with Risk Analysts in Indian banks, and used those insights to create high-fidelity sketch mockups.

FELLOWSHIPS, HONORS AND AWARDS

CMLH Generative AI in Healthcare Fellowship, Carnegie Mellon University (2024)

Awarded a \$100,000 fellowship supporting research on generative AI applications in healthcare and human-centered design.

ICoRD'21 2021 - Most Distinguished Paper

Development & Evaluation of Usability Heuristics for Voice User Interfaces.

IndiaHCI 2018 - Winner, Student Design Consortium

First prize at the IndiaHCI 2018 Student Design Consortium.

TALKS AND PRESENTATIONS

Invited talk, Memorial Sloan Kettering Cancer Center

Jun 2026

Input × Output: Mapping the Interaction Modality Space in Conversational Visual Analytics.

Invited Talk, MoSHI Lab Meeting, University of Pittsburgh

Mar 2026

Exploring Patient Interactions with AI Risk Stratification in Chronic Care.

Poster Presentation, CHI '25 Workshop on Envisioning the Future of Interactive Health

May 2025

Clini-Compare: An Interactive Patient-Similarity Visualization Tool for Clinical Decision Support.

TEACHING

Carnegie Mellon University

Pittsburgh, PA

Graduate Teaching Assistant, User Centered Research and Evaluation

Aug 2026 - Dec 2026

- Delivered lectures and led in-class research activities teaching user research methods, including interviewing, contextual inquiry, and data analysis.
- Graded student assignments for the course.

The MathWorks

Atlanta, GA

Lead Instructor

Aug 2023

- Led the Try Design Workshop, a two day event to teach design methods to a multi-disciplinary group of contributors.

Georgia Institute of Technology

Atlanta, GA

Graduate Teaching Assistant, Data Visualization

Aug 2020 - Dec 2020

- Facilitated remote learning by leading online lab sessions, virtual office hours, and offering project support via discussion boards and video calls.
- Developed additional online resources and recorded tutorials to help students with d3.js visualizations in a fully remote setting.

Georgia Institute of Technology

Atlanta, GA

Graduate Teaching Assistant, Data Visualization

Jan 2020 - May 2020

- Provided in-person support, guiding students through projects using d3.js and assisting with lab sessions, office hours, and project evaluations.
- Collaborated with the instructor to create course materials and grading rubrics, ensuring fair and consistent assessments.

MENTORSHIP

Isabella Hu

Aug 2026 - Present

Undergraduate Student, Carnegie Mellon University

Marissa Lee

Aug 2026 - Present

Undergraduate Student, Carnegie Mellon University

Grace Sng

Aug 2026 - Present

Undergraduate Student, Carnegie Mellon University

Alexander Palomino

May 2026 - Aug 2026

Undergraduate Student, Stony Brook University

Research Assistant through Research Experiences for Undergraduates (REU)

Anissa Patel

Jan 2026 - Aug 2026

MS-HCI Student, Carnegie Mellon University

Jonathan Lai

May 2025 - May 2026

Undergraduate Student, Carnegie Mellon University

Now MS-CS Student at Columbia University

Udhirna Krishnamurthy

Jan 2025 - Dec 2025

Undergraduate Student, Carnegie Mellon University

SERVICE

Program Committee

IEEE VIS 2026 Workshop on Visualization for AI Explainability (VISxAI)

Reviewer

ACM CHI 2027 (Papers)

IEEE VIS 2026 (Full Papers, Short Papers)

SKILLS

Research Methods: Systematic Literature Reviews, Ethnographic Observation, Contextual Inquiry, Think-Aloud Studies, Semi-Structured Interviews, Self-Assessment Surveys, Thematic Analysis

Programming Languages: Python, Javascript, Java, C++, C#

Techniques: Deep Learning, Data Visualization, NLP, UI Design, Computer Graphics